7

第八周作业

习题 7.1: P223 Ex 5.5.3

设数域$\mathbb{F}$上n维线性空间V是子空间U与W的直和，则对任意$\alpha \in V$，存在唯一一对向量β与$\gamma, \beta \in U, \gamma \in W$，使得$\alpha = \beta + \gamma$. 定义映射$\mathscr{A} : V \to V$如下：设$\alpha \in V$，则令$\mathscr{A}(\alpha) = \beta$. 映射$\mathscr{A}$称为$V$沿子空间$W$在$U$上的投影变换。证明：

(1) 投影变换$\mathscr{A}$是线性变换；

(2) 线性变换$\mathscr{B} : V \to V$为投影变换的充分必要条件是$\mathscr{B}$为幂等变换，即$\mathscr{B}$满足$\mathscr{B}^2 = \mathscr{B}$；

(3) 线性变换$\mathscr{B} : V \to V$为投影变换的充分必要条件是$\mathscr{I} - \mathscr{B}$为投影变换，其中$\mathscr{I}$是单位映射。

证明

(1) 设$\alpha_1 = \beta_1 + \gamma_1, \alpha_2 = \beta_2 + \gamma_2$为$V$中任意两个向量，其中$\beta_i \in U, \gamma_i \in W, i = 1, 2$. 对任意$\lambda, \mu \in \mathbb{F}$，由$U, W$是子空间及分解的唯一性，有$\mathscr{A}(\lambda\alpha_1 + \mu\alpha_2) = \lambda\beta_1 + \mu\beta_2 = \lambda\mathscr{A}(\alpha_1) + \mu\mathscr{A}(\alpha_2)$. 故$\mathscr{A}$线性。

(2) 若$\mathscr{B}^2 = \mathscr{B}$，则令$U = \mathrm{Im}(\mathscr{B}), W = \mathrm{Ker}(\mathscr{B})$. 若$\beta \in U \cap W$，则存在$\alpha \in V$使得$\beta = \mathscr{B}(\alpha)$，故$\beta = \mathscr{B}^2(\alpha) = \mathscr{B}(\beta) = 0$. 因此$U \cap W = \{0\}$. 对任意$\alpha \in V$，有$\alpha = \mathscr{B}(\alpha) + (\alpha - \mathscr{B}(\alpha)) = \beta + \gamma$，其中$\beta = \mathscr{B}(\alpha) \in U$，且$\mathscr{B}(\gamma) = \mathscr{B}(\alpha) - \mathscr{B}^2(\alpha) = 0$，所以$\gamma \in W$. 故$V = U \oplus W$，且$\mathscr{B}(\alpha) = \beta$，即$\mathscr{B}$为$V$沿子空间$W$在$U$上的投影变换。

反之若$\mathscr{B}$为投影变换，则存在V的子空间U, W满足$V = U \oplus W$，且对$\alpha = \beta + \gamma$，其中$\beta \in U, \gamma \in W$，有$\mathscr{B}(\alpha) = \beta$. 又由$\beta \in U$可知$\mathscr{B}^2(\alpha) = \mathscr{B}(\beta) = \beta = \mathscr{B}(\alpha)$，即$\mathscr{B}^2 = \mathscr{B}$.

(3) 由(2)知只需证$\mathscr{B}$为幂等变换当且仅当$\mathscr{I} - \mathscr{B}$为幂等变换。若$\mathscr{B}^2 = \mathscr{B}$，则$(\mathscr{I} - \mathscr{B})^2 = \mathscr{I} - 2\mathscr{B} + \mathscr{B}^2 = \mathscr{I} - \mathscr{B}$. 反之若$\mathscr{I} - \mathscr{B}$为幂等变换，对$\mathscr{I} - \mathscr{B}$应用同一结论即有$\mathscr{B} = \mathscr{I} - (\mathscr{I} - \mathscr{B})$为幂等变换。 □

习题 7.2: P223 Ex 5.5.8

设$\mathscr{A} : V \to V$与$\mathscr{B} : V \to V$是幂等变换，即$\mathscr{A}^2 = \mathscr{A}, \mathscr{B}^2 = \mathscr{B}$. 证明：

(1) $\mathrm{Im}(\mathscr{A}) = \mathrm{Im}(\mathscr{B})$的充分必要条件是$\mathscr{A}\mathscr{B} = \mathscr{B}, \mathscr{B}\mathscr{A} = \mathscr{A}$；

(2) $\mathrm{Ker}(\mathscr{A}) = \mathrm{Ker}(\mathscr{B})$的充分必要条件是$\mathscr{A}\mathscr{B} = \mathscr{A}, \mathscr{B}\mathscr{A} = \mathscr{B}$.

证明

(1) 由$\mathscr{A}\mathscr{B}=\mathscr{B}$可知$\mathrm{Im}(\mathscr{B})\subseteq\mathrm{Im}(\mathscr{A})$，由$\mathscr{B}\mathscr{A}=\mathscr{A}$可知$\mathrm{Im}(\mathscr{A})\subseteq\mathrm{Im}(\mathscr{B})$. 因此$\mathrm{Im}(\mathscr{A})=\mathrm{Im}(\mathscr{B})$.
反之，对任意$\alpha\in V$，由$\mathscr{B}(\alpha)\in\mathrm{Im}(\mathscr{B})=\mathrm{Im}(\mathscr{A})$知存在$\beta\in V$使得$\mathscr{B}(\alpha)=\mathscr{A}(\beta)$. 故$\mathscr{B}(\alpha)=\mathscr{A}(\beta)=\mathscr{A}^2(\beta)=\mathscr{A}(\mathscr{A}(\beta))=\mathscr{A}(\mathscr{B}(\alpha))=(\mathscr{A}\mathscr{B})(\alpha)$，即$\mathscr{A}\mathscr{B}=\mathscr{B}$. 类似地，由$\mathscr{A}(\alpha)\in\mathrm{Im}(\mathscr{B})$，可取$\gamma\in V$使得$\mathscr{A}(\alpha)=\mathscr{B}(\gamma)$，从而$\mathscr{A}(\alpha)=\mathscr{B}(\gamma)=\mathscr{B}^2(\gamma)=\mathscr{B}(\mathscr{B}(\gamma))=\mathscr{B}(\mathscr{A}(\alpha))=(\mathscr{B}\mathscr{A})(\alpha)$. 故$\mathscr{B}\mathscr{A}=\mathscr{A}$.
(2) 由$\mathscr{A}\mathscr{B}=\mathscr{A}$可知$\mathrm{Ker}(\mathscr{B})\subseteq\mathrm{Ker}(\mathscr{A})$，由$\mathscr{B}\mathscr{A}=\mathscr{B}$可知$\mathrm{Ker}(\mathscr{A})\subseteq\mathrm{Ker}(\mathscr{B})$. 因此$\mathrm{Ker}(\mathscr{A})=\mathrm{Ker}(\mathscr{B})$.
反之，对任意$\alpha\in V$，有$\alpha=\mathscr{B}(\alpha)+(\alpha-\mathscr{B}(\alpha))=\beta+\gamma$，其中$\beta=\mathscr{B}(\alpha)$. 由于$\mathscr{B}^2=\mathscr{B}$，故$\gamma\in\mathrm{Ker}(\mathscr{B})=\mathrm{Ker}(\mathscr{A})$. 于是有$\mathscr{A}(\alpha)=\mathscr{A}(\beta+\gamma)=\mathscr{A}\mathscr{B}(\alpha)$，即$\mathscr{A}=\mathscr{A}\mathscr{B}$. 类似地有$\alpha-\mathscr{A}(\alpha)\in\mathrm{Ker}(\mathscr{A})=\mathrm{Ker}(\mathscr{B})$，故$\mathscr{B}(\alpha)=\mathscr{B}\mathscr{A}(\alpha)$，即$\mathscr{B}=\mathscr{B}\mathscr{A}$. □

习题 7.3: P224 Ex 5.5.13

设$A\in\mathbb{C}^{n\times n},f(x),g(x)\in\mathbb{C}[x]$，且$(f(\lambda),g(\lambda))=1$. 证明：

$$\mathrm{rank}\,f(A)+\mathrm{rank}\,g(A)=n+\mathrm{rank}\,f(A)g(A).$$

证明

记$V=\mathbb{C}^n$，取线性变换$\mathscr{A}:V\to V,\alpha\mapsto A\alpha$. 则$\mathrm{rank}\,f(A)=\dim\mathrm{Im}(f(\mathscr{A}))$, $\mathrm{rank}\,g(A)=\dim\mathrm{Im}(g(\mathscr{A}))$. 由$f(x),g(x)$互素，故存在$u(x),v(x)\in\mathbb{C}[x]$使得$u(x)f(x)+v(x)g(x)=1$，从而有$\mathscr{I}=f(\mathscr{A})u(\mathscr{A})+g(\mathscr{A})v(\mathscr{A})$. 因此对任意$\alpha\in V$都有$\alpha=f(\mathscr{A})u(\mathscr{A})(\alpha)+g(\mathscr{A})v(\mathscr{A})(\alpha)\in\mathrm{Im}(f(\mathscr{A}))+\mathrm{Im}(g(\mathscr{A}))$. 从而有$V=\mathrm{Im}(f(\mathscr{A}))+\mathrm{Im}(g(\mathscr{A}))$. 由维数定理可知

$$\mathrm{rank}\,f(A)+\mathrm{rank}\,g(A)=n+\dim(\mathrm{Im}(f(\mathscr{A}))\cap\mathrm{Im}(g(\mathscr{A}))).$$

故只需证$\mathrm{Im}(f(\mathscr{A}))\cap\mathrm{Im}(g(\mathscr{A}))=\mathrm{Im}(f(\mathscr{A})g(\mathscr{A}))$. 由$f(\mathscr{A})g(\mathscr{A})=g(\mathscr{A})f(\mathscr{A})$可知$\mathrm{Im}(f(\mathscr{A})g(\mathscr{A}))\subseteq\mathrm{Im}(f(\mathscr{A}))\cap\mathrm{Im}(g(\mathscr{A}))$. 反过来，若$\alpha\in\mathrm{Im}(f(\mathscr{A}))\cap\mathrm{Im}(g(\mathscr{A}))$，则存在$\beta,\gamma\in V$使得$\alpha=f(\mathscr{A})(\beta)=g(\mathscr{A})(\gamma)$，于是$\alpha=f(\mathscr{A})u(\mathscr{A})(g(\mathscr{A})(\gamma))+g(\mathscr{A})v(\mathscr{A})(f(\mathscr{A})(\beta))=f(\mathscr{A})g(\mathscr{A})(u(\mathscr{A})(\gamma)+v(\mathscr{A})(\beta))\in\mathrm{Im}(f(\mathscr{A})g(\mathscr{A}))$. 故两者相等，代入上面的维数等式即得结论。 □

习题 7.4: P229 Ex 5.6.3

设$\mathscr{A}:V\to V$是线性变换，而$0\neq\alpha\in V$，证明：V中向量$\alpha,\mathscr{A}(\alpha),\cdots,\mathscr{A}^k(\alpha),\cdots$生成的子空间$U$是$\mathscr{A}$的不变子空间。设$\dim U=r$，证明：$\{\alpha,\mathscr{A}(\alpha),\cdots,\mathscr{A}^{r-1}(\alpha)\}$是$U$的基。求$\mathscr{A}|_U$在这组基下的方阵。

证明

如果$\beta\in U$，那么β是$\alpha,\mathscr{A}(\alpha),\cdots$的有限线性组合。由$\mathscr{A}$的线性性可知$\mathscr{A}(\beta)$也是这些向量的有限线性组合，故$\mathscr{A}(\beta)\in U$. 因此$U$是$\mathscr{A}$的不变子空间。
若$\dim U=r$，下证$\alpha,\mathscr{A}(\alpha),\cdots,\mathscr{A}^{r-1}(\alpha)$线性无关。否则，由$\alpha\neq0$，存在最小的正整数$m\leqslant r-1$，使得$\mathscr{A}^m(\alpha)$是$\alpha,\mathscr{A}(\alpha),\cdots,\mathscr{A}^{m-1}(\alpha)$的线性组合。写为$\mathscr{A}^m(\alpha)=c_0\alpha+c_1\mathscr{A}(\alpha)+\cdots+c_{m-1}\mathscr{A}^{m-1}(\alpha)$，其中$c_0,c_1,\cdots,c_{m-1}\in\mathbb{F}$. 对这一等式反复作用$\mathscr{A}$可知对任意$k\geqslant m$, $\mathscr{A}^k(\alpha)$都可由这m个向量线性表示。因此$\dim U\leqslant m\leqslant r-1$，与$\dim U=r$矛盾。故这$r$个向量线性无关，因而构成$U$的一组基。设$\mathscr{A}^r(\alpha)=a_0\alpha+a_1\mathscr{A}(\alpha)+\cdots+a_{r-1}\mathscr{A}^{r-1}(\alpha)$，其中$a_0,a_1,\cdots,a_{r-1}\in\mathbb{F}$. 于是$\mathscr{A}|_U$在基$\{\alpha,\mathscr{A}(\alpha),\cdots,\mathscr{A}^{r-1}(\alpha)\}$

下的矩阵为

$$\begin{pmatrix} 0 & 0 & \cdots & 0 & a_0 \\ 1 & 0 & \cdots & 0 & a_1 \\ 0 & 1 & \cdots & 0 & a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & a_{r-1} \end{pmatrix}.$$

当$r=1$时，此矩阵为(a_0). □

习题 7.5: P229 Ex 5.6.5

设V是n维复线性空间，且$\mathscr{A}: V \to V$是线性变换。证明：V的每个子空间都是$\mathscr{A}$的不变子空间的充分必要条件是$\mathscr{A}$为纯量变换，即$\mathscr{A}=\lambda\mathscr{I}$，其中$\lambda \in \mathbb{C}$.

证明

充分性：若$\mathscr{A}=\lambda\mathscr{I}$，则对任意子空间$U \subseteq V$，有$\mathscr{A}(U)=\lambda U \subseteq U$.

必要性：固定V的一组基$\{\alpha_1, \alpha_2, \cdots, \alpha_n\}$. 由假设知对每个$1 \leqslant j \leqslant n$，一维子空间$V_j=\operatorname{span}\{\alpha_j\}$是$\mathscr{A}$的不变子空间。因此存在$\lambda_j \in \mathbb{C}$使得$\mathscr{A}(\alpha_j)=\lambda_j\alpha_j$. 下面证明$\lambda_1=\cdots=\lambda_n$. 对任意$i \neq j$，考虑子空间$U=\operatorname{span}\{\alpha_i+\alpha_j\}$. 由于$U$是$\mathscr{A}$的不变子空间，存在$\lambda \in \mathbb{C}$使得$\mathscr{A}(\alpha_i+\alpha_j)=\lambda_i\alpha_i+\lambda_j\alpha_j=\lambda(\alpha_i+\alpha_j)$. 由$\alpha_i, \alpha_j$线性无关可知$\lambda_i=\lambda_j=\lambda$. 因此$\lambda_1=\cdots=\lambda_n$. 取$\lambda=\lambda_1$，即得$\mathscr{A}=\lambda\mathscr{I}$. □

习题 7.6: P239 Ex 5.7.7

设A与B为n阶复方阵，且方阵A的特征多项式为$\varphi(\lambda)$. 证明：方阵$\varphi(B)$可逆的充分必要条件是，方阵A与B没有公共特征值。

证明

设$\varphi(\lambda)=(\lambda-\lambda_1)\cdots(\lambda-\lambda_n)$. 其中$\lambda_1, \cdots, \lambda_n \in \mathbb{C}$. 于是

$$\det \varphi(B)=\det((B-\lambda_1 I_n)\cdots(B-\lambda_n I_n))=\prod_{i=1}^{n}\det(B-\lambda_i I_n).$$

如果存在λ_i是A和B的公共特征值，那么$\det(B-\lambda_i I_n)=0$，从而$\det\varphi(B)=0$，即$\varphi(B)$不可逆。

反之若A和B没有公共特征值，则对每个i, $\det(B-\lambda_i I_n) \neq 0$. 故$\det\varphi(B) \neq 0$，即$\varphi(B)$可逆。 □

习题 7.7: P239 Ex 5.7.11

证明：准对角方阵的最小多项式等于每个对角块的最小多项式的最小公倍式。

证明

设$A=\operatorname{diag}(A_1, \cdots, A_k)$，且$d_j(\lambda)$是$A_j$的最小多项式$(j=1, \cdots, k)$. 记$A$的最小多项式为$d_A(\lambda)$，并令$p(\lambda)=\operatorname{lcm}(d_1(\lambda), \cdots, d_k(\lambda))$. 对每个$j$，存在$c_j(\lambda)$使得$p(\lambda)=c_j(\lambda)d_j(\lambda)$，因此

$$p(A)=\operatorname{diag}(c_1(A_1)d_1(A_1), \cdots, c_k(A_k)d_k(A_k))=0.$$

故$p(\lambda)$是A的化零多项式，从而$d_A(\lambda) \mid p(\lambda)$. 而$d_A(A)=\operatorname{diag}(d_A(A_1), \cdots, d_A(A_k))=0$，故$d_A(A_j)=0, j=1, \cdots, k$. 故$d_j(\lambda) \mid d_A(\lambda), j=1, \cdots, k$，则$p(\lambda) \mid d_A(\lambda)$. 由$p(\lambda)$, $d_A(\lambda)$均首一知$p(\lambda)=d_A(\lambda)$. □

习题 7.8: P239 Ex 5.7.13(1)

求下列方阵的特征多项式与最小多项式：

$$(1)\begin{pmatrix} 0 & 1 & & & \\ & 0 & 1 & & \\ & & \ddots & \ddots & \\ & & & 0 & 1 \\ a_0 & a_1 & & \cdots & a_{n-1} \end{pmatrix}_{n\times n}.$$

解答

记题中矩阵为A. 将行列式$\det(\lambda I_n - A)$按最后一行展开得到$\varphi_A(\lambda) = \lambda^n - a_{n-1}\lambda^{n-1} - \cdots - a_1\lambda - a_0$. 下求最小多项式$d(\lambda)$. 设$e_i$为仅第$i$个元素为1，其余为零的行向量，直接计算得$e_1A = e_2$, $e_1A^2 = e_2A = e_3, \cdots, e_1A^{n-1} = e_{n-1}A = e_n$, $e_1A^n = e_nA = (a_0, \cdots, a_{n-1})$. 若$\deg d < n$，则存在$b_0, \cdots, b_{n-1} \in \mathbb{C}$不全为0使得$d(\lambda) = b_0 + b_1\lambda + \cdots + b_{n-1}\lambda^{n-1}$. 由$d(A) = 0$知$0 = e_1d(A) = b_0e_1 + b_1e_1A + \cdots + b_{n-1}e_1A^{n-1} = b_0e_1 + b_1e_2 + \cdots + b_{n-1}e_n$. 则只有$b_0 = \cdots = b_{n-1} = 0$，矛盾。从而$\deg d = n$，故$d(\lambda) = \varphi_A(\lambda)$.

习题 7.9: P240 Ex 5.7.16

取方阵$A \in \mathbb{C}^{n\times n}$. 定义映射$\mathscr{A} : \mathbb{C}^{n\times n} \to \mathbb{C}^{n\times n}$如下：设$X \in \mathbb{C}^{n\times n}$，则令$\mathscr{A}(X) = AX$. 证明：线性映射$\mathscr{A}$的最小多项式等于方阵$A$的最小多项式。

证明

按列排列标准矩阵单位$E_{ij}, 1 \leqslant i, j \leqslant n$，作为$\mathbb{C}^{n\times n}$的一组基。由$AE_{ij} = \sum_{k=1}^{n} a_{ki}E_{kj}$可知$\mathscr{A}$在这组基下的矩阵为$\operatorname{diag}(A, \cdots, A)$，共有$n$个对角块，每个对角块均为$A$. 由Ex 5.7.11可知$\operatorname{diag}(A, \cdots, A)$的最小多项式等于$A$的最小多项式。又因为线性映射的最小多项式等于它在任意一组基下的矩阵的最小多项式，所以要证的结论成立。 □

习题 7.10: P240 Ex 5.7.20

设n阶方阵A为

$$A = \begin{pmatrix} a_1(1-a_1) & -a_1a_2 & \cdots & -a_1a_n \\ -a_2a_1 & a_2(1-a_2) & \cdots & -a_2a_n \\ \vdots & \vdots & \ddots & \vdots \\ -a_na_1 & -a_na_2 & \cdots & a_n(1-a_n) \end{pmatrix}.$$

当$a_1, a_2, \cdots, a_n$满足什么条件时方阵A可逆，并当A可逆时，求逆方阵A^{-1}.

解答

记$s = a_1 + a_2 + \cdots + a_n$. 利用加边法可以得到：

$$\det A = a_1a_2\cdots a_n \begin{vmatrix} 1-a_1 & -a_1 & \cdots & -a_1 \\ -a_2 & 1-a_2 & \cdots & -a_2 \\ \vdots & \vdots & \ddots & \vdots \\ -a_n & -a_n & \cdots & 1-a_n \end{vmatrix} = a_1a_2\cdots a_n(1-s).$$

因此A可逆当且仅当$a_1a_2\cdots a_n\neq 0$且$a_1+a_2+\cdots+a_n\neq 1$.

记$\Lambda=\mathrm{diag}(a_1,a_2,\cdots,a_n)$, $a=(a_1,a_2,\cdots,a_n)^\top$,则$A=\Lambda-aa^\top$. 由$A$可逆知$\Lambda$可逆。令$e=(1,\cdots,1)^\top$,则$a=\Lambda e$，故$A=\Lambda(I_n-ea^\top)$. 令$B=ea^\top$，则$B^2=e(a^\top e)a^\top=sB$. 从而$B^k=s^{k-1}B, k\geqslant 1$. 结合泰勒展开$(1-x)^{-1}=1+x+x^2+\cdots$可以验证$(I_n-B)^{-1}=I_n+(1-s)^{-1}B$. (由$A$可逆有$s\neq 1$). 于是

$$A^{-1}=(I_n-ea^\top)^{-1}\Lambda^{-1}=\big(I_n+(1-s)^{-1}ea^\top\big)\Lambda^{-1}=\Lambda^{-1}+(1-s)^{-1}ea^\top\Lambda^{-1}.$$

由于$a^\top\Lambda^{-1}=e^\top$，所以$A^{-1}=\mathrm{diag}(a_1^{-1},a_2^{-1},\cdots,a_n^{-1})+(1-a_1-a_2-\cdots-a_n)^{-1}J$，其中$J$是所有元素均为1的$n$阶方阵。